Constantinos Antoniou Research Profile

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At a glance

Prof. Constantinos Antoniou has produced 533 indexed research works since 1997, cited about 11,800 times, with an h-index of 55 in OpenAlex (53 in Scopus, 69 in Google Scholar). Two thirds of that output dates from his decade at TUM, and 2025 and 2026 are the two most productive years of his career.

Measure	OpenAlex	Scopus	Google Scholar
Works indexed	533	380+	500+
Citations	11,785	10,000+	17,000+
h-index	55	53	69
i10-index (works with 10+ citations)	224	–	206
Works with 100+ citations	24	–	30

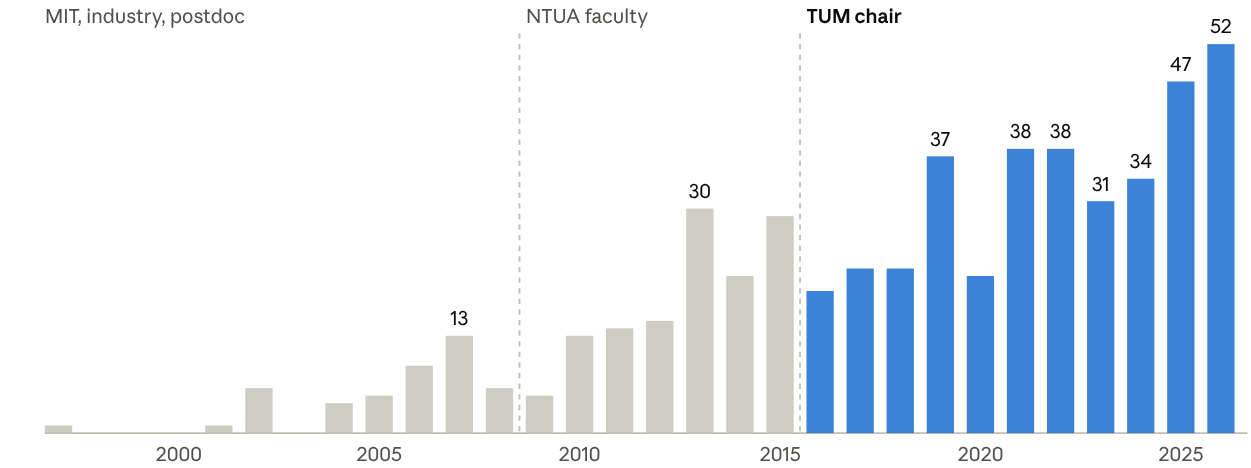
Beyond the citation counts: 550+ distinct co-authors across 150+ institutions in 40 countries; co-authorship on 24 papers with over 100 citations each; 58% of works open access; 9 EU-funded projects as PI and over €4 million in third-party funding in the last five years; Co-Editor-in-Chief of Transportation Research Part A; 7 doctorates completed and 17+ in progress. Scopus and Scholar figures come from the chair's own profile text; OpenAlex figures were extracted on 3 October 2026.

Three decades, three gears

Productivity has stepped up with each career stage rather than drifting: 45 works in the MIT and industry years (1997 to 2008, about 4 a year), 127 in seven NTUA faculty years (about 18 a year), and 361 in the TUM decade so far (about 33 a year).

Output rose from about 4 works a year at MIT to about 33 a year at TUM

Indexed works per year, 1997 to 2026 (2026 is year to date); TUM years in blue



533 works in total; cumulative count passed 100 in 2013, 300 in 2021 and 500 in 2026

OpenAlex counts_by_year for author A5001459918, extracted 3 Oct 2026 · 30 years

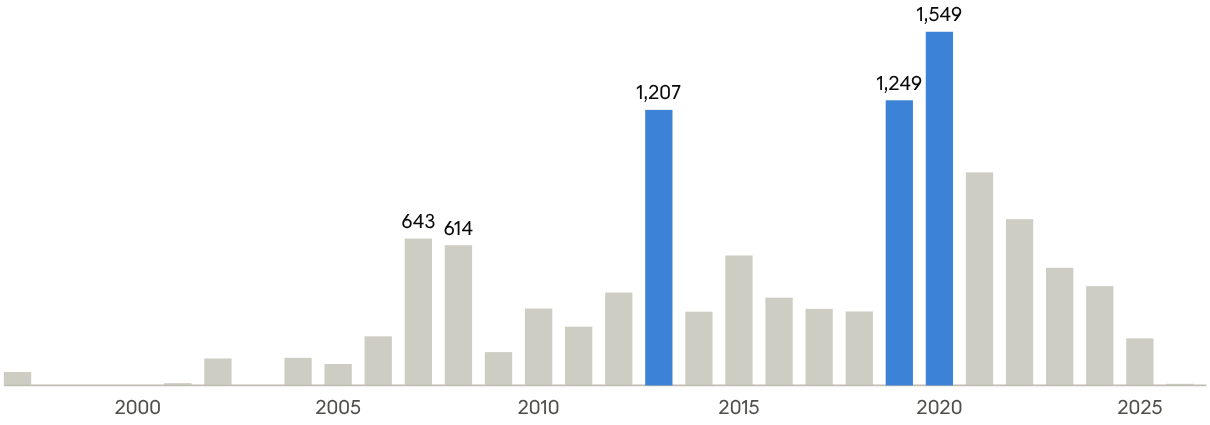
The NTUA era peaks in 2013 (30 works), the first year above 20. At TUM the floor itself moved: no year since 2016 has fallen below 19, and the last six years average 40. The 2026 count (52 by early October) already exceeds every previous full year, driven partly by the growth of the chair's doctoral group, which now exceeds 17 students.

Impact shape

Impact is concentrated in a few landmark papers but rests on a broad base: 407 of 533 works have been cited at least once, 224 ten or more times, and the top 25 papers alone account for 42% of all citations. The citation record has three peaks, each marking a different research agenda.

Three citation peaks: 2007, 2013 and 2019 to 2020

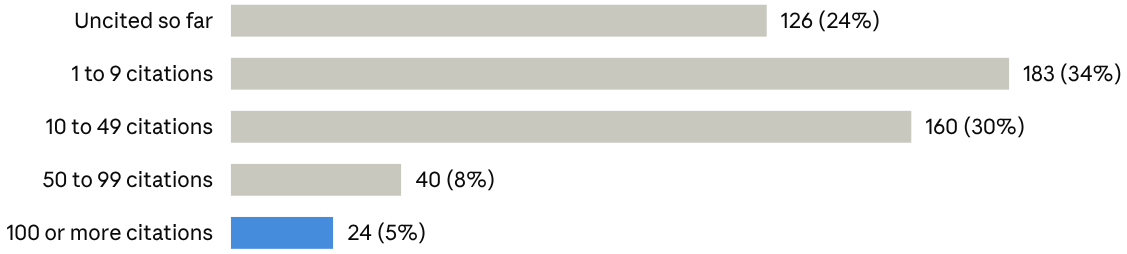
Citations received to date by the works published in each year; years above 1,000 in blue



11,785 citations in total; recent years are still accruing

224 works have ten or more citations; 24 have a hundred or more

Number of works in each citation tier (median work: 5 citations, mean: 22)



533 works in total; the 25 most-cited carry 42% of all citations

OpenAlex works for author A5001459918, grouped by publication year and by cited_by_count, extracted 3 Oct 2026

The 2007 peak is methodological (calibration of dynamic traffic assignment and microscopic simulation models); 2013 is the NTUA policy-and-safety peak (vehicle sharing adoption, weather and accident risk, infrastructure and house prices); 2019 to 2020 is the TUM future-mobility peak (urban air mobility, shared autonomous vehicles, machine-learning traffic forecasting). The ten most-cited works span all three agendas.

Rank	Work	Year	Citations
1	Shared autonomous vehicle services: a comprehensive review	2020	700
2	Public transit user satisfaction: variability and policy implications	2008	526

Rank	Work	Year	Citations
3	Factors affecting the adoption and use of urban air mobility	2019	345
4	Factors affecting the adoption of vehicle sharing systems by young drivers	2013	326
5	Explaining the road accident risk: weather effects	2013	293
6	Exploring preferences for transportation modes in an urban air mobility environment: Munich case study	2019	225
7	How do transport infrastructure and policies affect house prices and rents? Evidence from Athens	2013	212
8	Nonlinear Kalman filtering algorithms for on-line calibration of dynamic traffic assignment models	2007	192
9	Factors affecting modal choice in urban mobility	2012	162
10	Long-term application potential of urban air mobility complementing public transport: an upper Bavaria example	2020	157

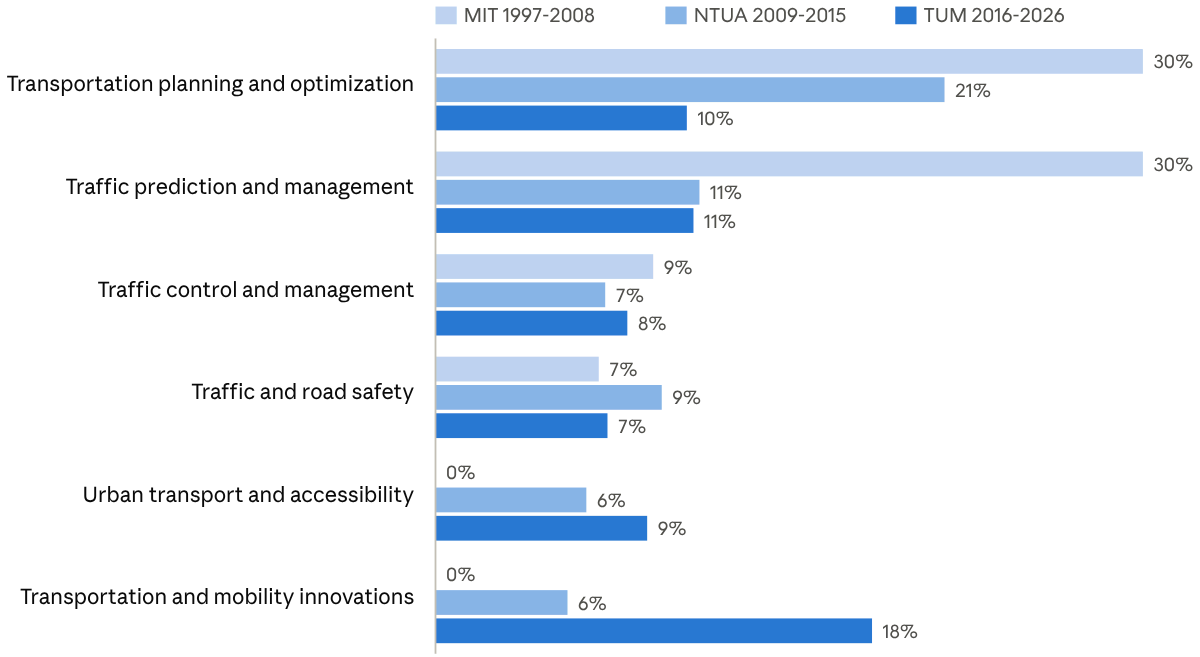
Citation counts are OpenAlex values on 3 October 2026; Google Scholar counts for the same papers run roughly 40 to 60% higher.

Topic migration

The research agenda has widened from one problem to a portfolio. At MIT, 26 of 44 works (59%) sat in just two topics, traffic prediction and transportation optimization; at TUM the two largest topics cover only 29% of output, and "Transportation and Mobility Innovations" has become the single largest, with 66 works since 2016.

Mobility innovations went from absent at MIT to the largest topic at TUM

Share of each era's works whose primary topic is the one named, in %; darker bar = later era



Six most frequent primary topics, 64% of all works; smaller topics make up the rest

OpenAlex primary_topic grouped by era for author A5001459918, extracted 3 Oct 2026 · 6 topics × 3 eras

The keyword record tells the same story in finer grain. The most frequent keywords across the whole career are road safety (57 works), dynamic traffic assignment (42), travel behavior (35), mode choice (30), microscopic traffic simulation (25) and driver behavior (24). The method vocabulary of the early years, online calibration (22), extended Kalman filter (12) and SPSA (12), sits alongside the TUM-era vocabulary of urban air mobility (21), car sharing (21), shared mobility (19), autonomous vehicles (18) and bike sharing (18). Two city names recur as study areas: Athens (17 works) and Munich (15).

What has not changed is the quantitative core: statistical and machine-learning methods, discrete choice and stated-preference surveys (19 works), and simulation are the common thread from the 2004 PhD on online calibration to the 2020s work on automated and shared mobility.

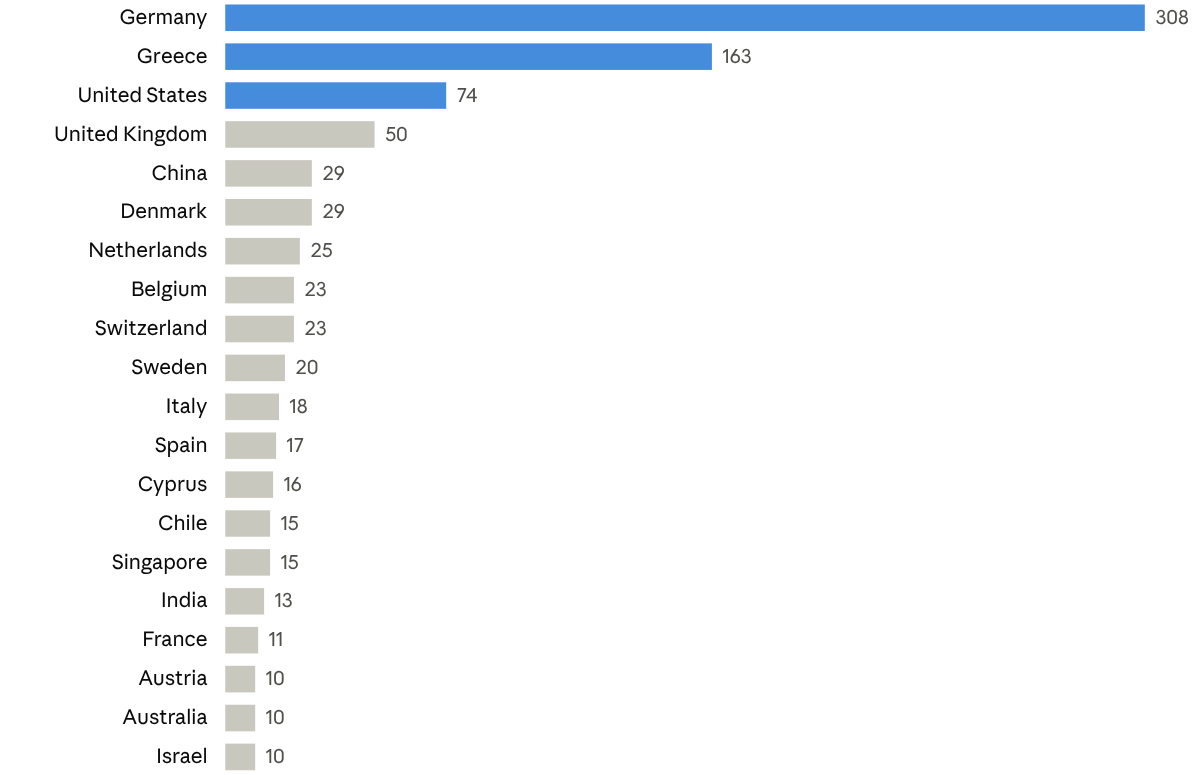
Seen through the UN Sustainable Development Goals, 149 works map to SDG 11 (sustainable cities and communities), 43 to SDG 9 (industry, innovation and infrastructure), 21 to SDG 3 (good health and well-being, mostly road safety) and 13 to SDG 13 (climate action).

Collaboration network

The network is large, international and visibly multi-generational: 550+ distinct co-authors, 150+ institutions, 40 countries, and a top co-author list that pairs his own doctoral advisors with his own doctoral students.

Co-authors from 40 countries: three home bases and a wide European tail

Works with at least one co-author affiliated in the country (a work can count in several); top 20 shown



Blue: countries of his own three career institutions (TUM, NTUA, MIT); a further 20 countries have 1 to 9 works each

OpenAlex works grouped by co-author institution country for author A5001459918, extracted 3 Oct 2026 · top 20 of 40 countries

Germany (308 works), Greece (163) and the United States (74) reflect the three career bases. The next tier is European: United Kingdom 50, Denmark 29, Netherlands 25, Belgium 23, Switzerland 23, Sweden 20. China (29), Chile (15), Singapore (15), India (13) and Australia (10) extend the reach beyond Europe. Since 2016 the mix has shifted further outward: Greece falls to 63 of 362 works while the UK, China, Denmark, the Netherlands and Belgium all keep or grow their share.

By institution, the most frequent partners after TUM (308) and NTUA (149) are MIT (43), UCL (28), DTU (27), Northeastern University (19), KTH (16), EPFL (14), Hasselt University (14), University of Cyprus (14), University of Chile (13) and TU Delft (13). Several of these

are where former group members now hold positions, so the collaboration graph doubles as an alumni map.

Co-author	Joint works	Relationship
George Yannis (NTUA)	54	NTUA colleague, road safety
Dimitrios Efthymiou	41	First PhD graduate (NTUA, 2014)
Moshe Ben-Akiva (MIT)	40	MS and PhD advisor
Haris Koutsopoulos (Northeastern)	36	PhD co-advisor
Christelle Al Haddad	35	TUM doctoral researcher, urban air mobility
Emmanouil Chaniotakis (UCL)	34	Former TUM group member
Mohamed Abouelela	31	TUM doctoral researcher, shared mobility
Santhanakrishnan Narayanan	26	TUM doctoral researcher, shared autonomous vehicles
Eleonora Papadimitriou (TU Delft)	24	NTUA road-safety collaborator
Qing-Long Lu	22	TUM doctoral researcher, traffic prediction

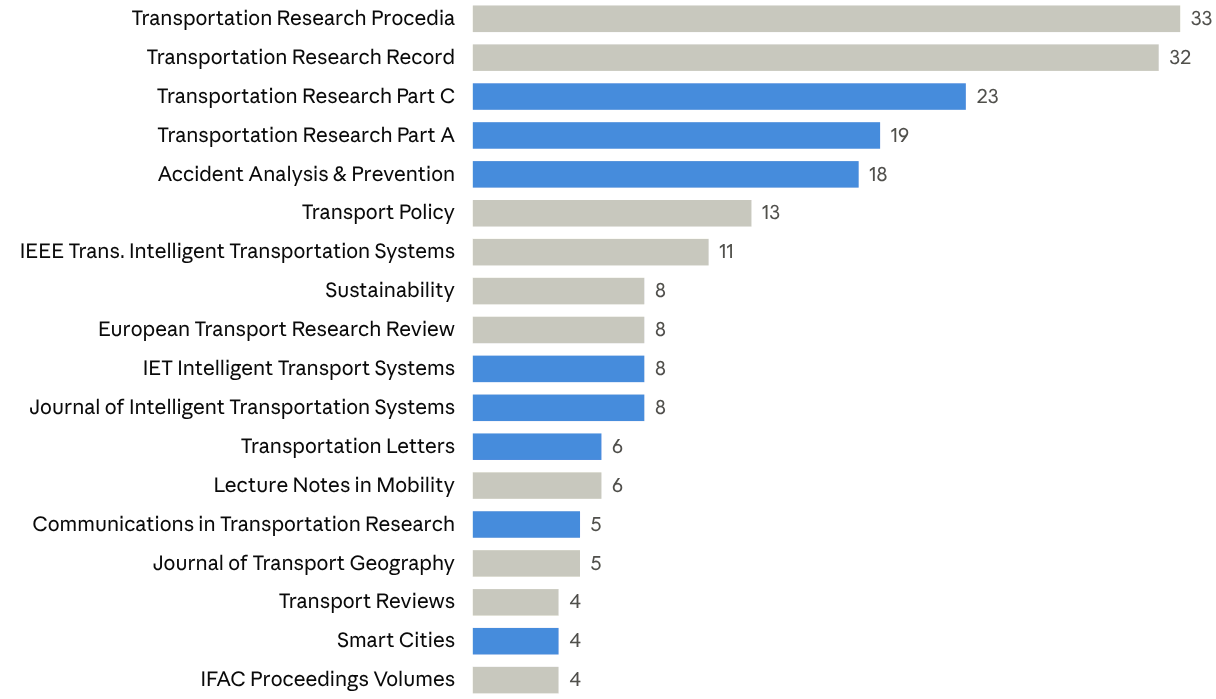
Team sizes are compact: the median paper has 4 authors, 70% have between 2 and 4, and only 5 are single-authored; 15 works have 10 or more authors, mostly EU-project consortia and two large horizon-scanning papers with over 50 authors. 309 of 533 works (58%) are open access.

Where the work appears

The venue list is the field's own shortlist. Eight of the top eighteen outlets are journals in which he serves or has served as editor, and the three Elsevier flagships, Transportation Research Parts A and C and Accident Analysis & Prevention, together hold 60 of his papers.

Published in 90+ journals; the top venues are the core transportation titles

Works per venue, top 18; blue = journals where he holds or held an editorial role



Preprint servers (SSRN 27, arXiv 6) and Elsevier eBooks (21 chapters) are omitted; TRB Annual Meeting compendia are split across years

OpenAlex works grouped by primary source for author A5001459918, extracted 3 Oct 2026 · top 18 venues

By publication type, OpenAlex counts 362 journal articles, 67 conference papers, 38 preprints, 32 book chapters, 8 editorials and 2 books; the chair's own Scopus count is 240+ journal articles, 80+ conference papers, 25+ book chapters and 4 books. TRB venues (Transportation Research Record plus the Annual Meeting compendia) and Transportation Research Procedia, which publishes selected papers from conferences such as hEART and mobil.TUM, account for the bulk of the conference stream.

Beyond papers

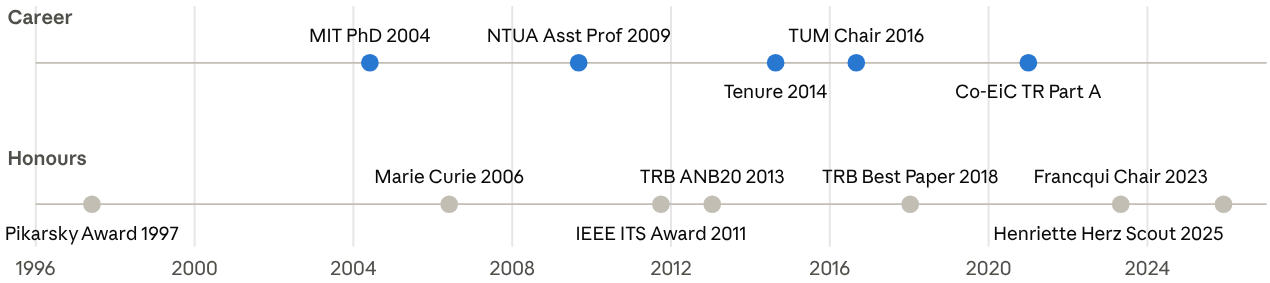
The publication record sits on top of a comparable record in funding, editing, supervision and service. In the last five years the chair has attracted over €4 million in third-party funding, and he has been principal investigator on nine EU-funded projects (SCUDO, CulturalRoad, MI-TRAP, AutoSUP, PHOEBE, i-DREAMS, MOMENTUM, Drive2theFuture, NOESIS) and five nationally funded ones (VOTAVI, SEE-FAr, D-Vanpool, TraMPA, PANAMERA). The three largest H2020 consortia he led a work package or partner role in had total budgets of €6.3 million (i-DREAMS), €4.0 million (Drive2theFuture) and €2.9 million (MOMENTUM).

Dimension	Figure
Third-party funding, last 5 years	€4 million+
EU-funded projects as PI	9
Nationally funded projects as PI	5
Editor-in-Chief roles	Co-EiC, Transportation Research Part A (with Anming Zhang)
Current editorial boards	TR Part C, Accident Analysis & Prevention, Communications in Transportation Research
Past editorial roles	Deputy EiC IET ITS; AE Transportation Letters; Editor EURO J. Transportation and Logistics
Doctorates completed / in progress	7 / 17+
Scientific committees served on	100+
Journals reviewed for	50+
Conferences organised	hEART 2018 (Athens), mobil.TUM 2018 and 2019, IEEE MT-ITS 2021, hEART 2025 (TUM), ISTTT26 (Munich, July 2026)
Programme directorships	MS Transportation Systems (TUM); MS Rail Transport and Logistics (TUM Asia, Singapore)

Alumni of the group now hold positions at UCL, DTU, UPC and the University of Arizona, among others, which is visible in the collaboration data above.

A recognition every few years, across three institutions and three decades

Career steps (top) and selected awards and honours (bottom), placed by date



Dates from the chair's website data and the 2020 CV · 12 milestones, 1997 to 2025

The honours run from the 1997 Milton Pikarsky Award for the best MS thesis in the US, through the 2011 IEEE ITS Outstanding Application Award for the DynaMIT dynamic traffic assignment system, to the 2023 Francqui Chair at UHasselt and the Humboldt Foundation's Henriette Herz Scout designation in December 2025.

Sources, method and caveats

All charts are drawn from the OpenAlex record for author A5001459918 (ORCID 0000-0003-0203-9542), extracted on 3 October 2026; headline Scopus and Google Scholar figures are quoted from the chair's own profile text. The three databases differ because they index different things: Scopus counts only peer-reviewed items in its journal list (380+ works), OpenAlex adds preprints, reports, datasets and editorials (533 works), and Google Scholar also counts theses, working papers and citations from non-indexed documents, which is why its citation total (17,000+) and h-index (69) run highest.

Three caveats for anyone reusing the numbers. OpenAlex year counts for 2025 and 2026 include preprints and SSRN postings that may later appear again as journal versions, so the recent surge is partly a double count that Scopus would not show. Country and institution counts are per work, and a work with co-authors in several countries counts once in each. The 2026 citation count (7) reflects publication recency, not impact.

- Chair profile, TUM School of Engineering and Design (metrics as of January 2025)
- Curriculum vitae, June 2020 (projects, teaching, service, awards)
- TUM Research Portal profile (385 outputs, 10,475 citations, fingerprint)
- OpenAlex author record and works API
- Google Scholar profile and Scopus author page
- Website Data.docx supplied by Rakib (timeline, Scopus-based metrics, 2023 to 2026 honours)